Lesson Plan • Geometry 6.4 Triangle Midsegment Theorem

Wed., Nov. 4 • 4th hour • 10:40 • Observer: Mr. Noble • Feedback on #9, #14, #15

Min	What	Notes
0–5	**Bell ringer (ACT)**	Midpoint of (−4, 7) and (10, −3). Answer: A (3, 2). B = subtracted, C = didn't divide by 2, D = sign error.
5–15	**Discovery in pairs**	One triangle sheet per pair. Circulate; make sure both partners write.
15–25	**Debrief + model**	Name it: DE joins two midpoints, so it is a midsegment. Then table + key, students state the rule, theorem, Ex. 1–3.
25–40	**Practice on Chromebooks**	Google Classroom → 6.4. Watch the Practice Results sheet; help low scores first.
40–45	**Closure + exit ticket**	Students finish: “A midsegment is ___ and ___.” Then hand out exit tickets. Collect them at the door.

Before class

- [ ] Print the discovery sheets (one triangle per pair) and the exit tickets (cut in half).
- [ ] Open the slides on the TV computer and click Present.
- [ ] Post the 6.4 practice link in 4th hour's Google Classroom (bonus is already included).
- [ ] Enter the bell ringer in edia.app.
- [ ] Have the Practice Results sheet open on your computer.

Worked examples (answers)

Ex. 1: BC = 26 → DE = 13. DE = 7.5 → BC = 15.

Ex. 2: DE = 3x − 2, BC = 4x + 6 → 2(3x − 2) = 4x + 6 → x = 5, DE = 13.

Ex. 3: A(−2, 4), B(−6, −4), C(4, −2): D(−4, 0), E(1, 1); both slopes 1/5; DE = √26, BC = 2√26.

Discovery Answer Key (teacher)

Every triangle: DE is parallel to BC (same slope) and half as long.

#	A	B	C	D	E	slope DE	slope BC	DE	BC
1	(2, 10)	(0, 0)	(6, 8)	(1, 5)	(4, 9)	4/3	4/3	5	10
2	(−4, 6)	(−6, −2)	(2, −8)	(−5, 2)	(−1, −1)	−3/4	−3/4	5	10
3	(4, −2)	(−4, 4)	(4, 10)	(0, 1)	(4, 4)	3/4	3/4	5	10
4	(−8, 0)	(0, 8)	(6, 0)	(−4, 4)	(−1, 0)	−4/3	−4/3	5	10
5	(0, −6)	(−8, 2)	(2, 2)	(−4, −2)	(1, −2)	0	0	5	10
6	(10, 6)	(2, −4)	(2, 8)	(6, 1)	(6, 7)	undefined	undefined	6	12

Early-finisher extension (F = midpoint of BC; DF vs. AC)

#1: F = (3, 4) DF = √5 AC = 2√5

#2: F = (−2, −5) DF = √58 AC = 2√58

#3: F = (0, 7) DF = 6 AC = 12

#4: F = (3, 4) DF = 7 AC = 14

#5: F = (−3, 2) DF = √17 AC = 2√17

#6: F = (2, 2) DF = √17 AC = 2√17

Debrief questions

- What did every group find about the slopes? What does equal slope tell us?
- What did every group find about the lengths?
- Triangle 5 has a horizontal side and triangle 6 a vertical one. Did the pattern still hold?
- Put it in one sentence: a midsegment is _____ to the third side and _____ as long.

Exit Ticket Answer Key (teacher)

1. DE = 9 (half of 18)
2. $2(2x + 1) = 5x - 4 \rightarrow 4x + 2 = 5x - 4 \rightarrow x = 6$. DE = 13 (BC = 26).
3. DE connects the midpoints of two sides of the triangle (AB and AC).

What wrong answers tell you

• #1 answered 36: doubled instead of halved. Mixed up which segment is shorter.
• #2 answered x = 5/3 (from 2x + 1 = 5x − 4): set them equal, forgot "half as long."
• #2 answered x = 6 but DE = 26: found x, then gave BC instead of DE.
• #3 "it's in the middle" or "it's parallel": describes a property, not the definition.

Sort for tomorrow

• 3/3: ready to move on.
• #1 right, #2 wrong: reteach setting up the equation with the factor of 2.
• #1 wrong: reteach the theorem itself with a picture before any algebra.